



Figure 2: Connection (Credits: <https://forum.huawei.com/enterprise/en/data/attachment/forum/202109/16/224428xxewr2e1l3to21zx.png>)

The receiver waits for the packets and assembles them based on the sequence number. Checksum is calculated on both the sending and receiving computer to make sure data integrity is maintained and see if the packet received is indeed the same as the packet that was sent.

Step 3: Connection termination

The session uses a four-way termination sequence by using an additional flag — the FINISHED (FIN) flag. Figure 3 explains the communication, although this happens only with a graceful connection release.

The protocols supported by TCP are:

- File Transfer Protocol (FTP)
- Hypertext Transfer Protocol (HTTP)
- Secure Shell (SSH)

UDP

The User Datagram Protocol, or UDP, is another widely used transport protocol. It's faster than TCP but is also less reliable. UDP does not make sure all packets are delivered and in order, and it doesn't establish a connection before beginning or receiving transmissions. UDP is a connectionless protocol, and the packet header size is also smaller than TCP.

Protocols supported by UDP are:

- Dynamic Host Transfer Protocol (DHCP)
- Domain Name System (DNS)
- Trivial File Transfer Protocol (TFTP)
- Voice over Internet Protocol (VoIP)

SSL/TLS

So far we have seen how communication happens in the transportation layer. But the challenge here is data security. The sniffed packets can be easily read. To overcome this, we have SSL/TLS to secure the data in the transport layer.

SSL stands for Secure Socket Layer, and ensures the data is encrypted. TLS stands for Transport Layer Security. It is the advanced version of SSL, but mostly still referred to as SSL. SSL 3.0 was deprecated by IETF in 2015.

SSL and TLS transport protocols are similar but not entirely interoperable. Here are some terms you should get familiar with in connection with these protocols.

Cipher suite: This is a set of algorithms used for securing communications and generally performs the following:

- Key exchange
- Encryption
- Authentication

Digital certificates: These are issued by a trusted certification authority or CA.

Encryption: This makes sure data is encoded in a way it can't be understood by the eavesdropping middleman. 256-bit encryption keys are now the minimum accepted level of secure connectivity for online banking or electronic commerce transactions.

Asymmetric keys: A public key can encrypt and a private key is used for decryption and signature verification. The common algorithm is RSA (Rivest Shamir Adleman).

Symmetric keys: These use single or shared keys for encryption and decryption.

Asymmetric encryption is not used for bulk data encryption as it is slower; it is used to transfer the symmetric key between two parties, which is quicker and more efficient for encrypting data.

TLS uses strong encryption methods, such as AES. It can work without encryption altogether if only authentication is needed.

TLS 1.3 only supports the modified version of the Diffie-Hellman public-key cryptography algorithm to share a symmetric key between a client and the server. It no longer has the support of the RSA algorithm for the key exchange.

TLS has two layers of operation:

- **TLS Record Protocol:** This allows the client and server to communicate using some form of encryption algorithm (or without encryption, if desired).
- **TLS Handshake Protocol:** This allows the client and server to authenticate one another and exchange the encryption keys to be used during the session.

Figure 4 shows how TLS works. You must first establish the TCP/UDP connection and then use the TLS handshake to make a secure connection.

- **DH parameter:** The Diffie-Hellman (DH) algorithm uses exponential calculations to arrive at the same premaster secret. The server and client each provide a parameter for the calculation, and when combined they result in a different calculation on each side, with results that are equal.
- **Client hello:** The client sends a hello message with the protocol version, the client random, and a list of cipher suites.
- **Server hello:** The server replies with its SSL certificate, its selected cipher suite, and the server random.
- **Authentication:** The client verifies the server's SSL certificate with the certificate authority that issued it. This confirms that the server is what it says it is, and that the